# AI-Based Stock Market Prediction & Portfolio Optimization System

*Northgate Quantitative Research - Applied Machine Learning Capstone - Written report*

*Educational research and decision-support tool only. Outputs are not financial advice.*

**IMPORTANT - DATA PROVENANCE:** this build had no internet access to Yahoo Finance, FRED or Hugging Face, so every number below comes from a clearly labelled SYNTHETIC stand-in market (src/ingest.py, --source synthetic) with realistic stylised facts (GARCH volatility, fat tails, a market factor, earnings jumps, injected data defects). The pipeline code is the deliverable; the results demonstrate that it runs end to end and are NOT evidence about real markets. Re-run `make all` with data.source: yfinance to regenerate this report on real data.

## 1. Executive summary

This project delivers a layered, reproducible pipeline - ingestion, cleaning, features, eight forecasting models, news sentiment, mean-variance optimisation, a rule-based recommender and a Streamlit dashboard - that moves from raw market data to a risk-aware, explainable allocation. All validation is strictly chronological (70/15/15 by date, windows never straddle a split, scalers fit on train only).

| Acceptance criterion (PRD 14) | Status | Evidence |
|---|---|---|
| **Best forecaster beats random walk on MAE and direction (h=1)** | MET | LSTM: MAE 4.902 vs 4.904; direction 51.7% vs 50.6% |
| **Optimised portfolio Sharpe > equal-weight (test window)** | MET | 0.75 vs 0.69 (95% CI optimised -0.56 to 1.98) |
| **Sentiment effect measured and reported honestly** | REPORTED | delta MAE -0.0055, delta dir. acc. +1.73 pts |
| **Eight-model table populated on a leak-free held-out window** | YES | 3420.0 test samples per model |
| **From-scratch maths verified against libraries** | PASS | 15 checks |

The grading philosophy of the brief applies: a modest, correct, leakage-free result with honest analysis is the goal. Where an acceptance bar is not met, the report says so (Section 14).

## 2. Problem statement and scope

A price forecast has no economic meaning until it is converted into a risk-adjusted allocation and compared with a sensible benchmark. The system therefore treats forecasting and allocation as one pipeline: sentiment feeds the forecasting layer, forecasts feed the optimisation and recommendation layers. Non-goals (per the brief): intraday trading, live execution, derivatives, reinforcement-learning agents and any profitability claim.

Universe: AAPL, MSFT, JPM, XOM, JNJ, PG, NVDA, KO, CAT, HD; benchmark ^GSPC; VIX; sector proxy XLK; horizons [1, 5] trading days; target = forward log return of the adjusted close (price forecast = P_t x exp(prediction)).

## 3. Data and cleaning pipeline

Source of this run: synthetic. The cleaning function (src/clean.py) is ordered and idempotent: de-duplicate, align to the ^GSPC trading calendar, forward-fill gaps of up to three sessions (never back-fill; longer gaps are flagged), flag outliers with a rolling z-score AND the 1.5xIQR fence, keep genuine events (a volume spike marks an earnings-type move) and winsorise only confirmed one-day bad prints, then enforce the OHLC invariant (violating rows quarantined and re-imputed from the last valid session). Macro series are shifted by their publication lag before forward-filling so a figure is never used before it was released. Every change is logged to data/processed/cleaning_log.csv.

| Data-quality metric | Value |
|---|---|
| **rows in** | 38503 |
| **rows out** | 38506 |
| **duplicates removed** | 2 |
| **gap sessions filled** | 5 |
| **gaps longer than 3 flagged** | 0 |
| **outliers flagged** | 69 |
| **outliers kept as events** | 68 |
| **outliers winsorised** | 1 |
| **ohlc violations** | 1 |

Data contract (PRD 3.4): history ge 8 years = PASS; universe size 8 to 15 = PASS; zero missing days post clean = PASS; single trading calendar = PASS.

## 4. Feature engineering and leakage controls

| Family | Features (all computed from data at or before t) |
|---|---|
| **returns** | ret1, ret5, ret10, ret21 |
| **trend** | p_sma10, p_sma20, p_sma50, p_sma200, p_ema10, p_ema20, p_ema50, p_ema200, macd, macd_signal, macd_hist |
| **momentum** | rsi14, stoch_k, stoch_d, roc10 |
| **volatility** | vol21, atr14, bb_width |
| **volume** | vol_z, obv_z, vol_price_div |
| **calendar** | dow, month, turn_of_month |
| **macro_market** | vix, vix_chg, yield_spread, index_ret1, sector_ret1, cpi_yoy, unrate |
| **lagged** | ret1_lag1, ret1_lag2, ret1_lag3, ret1_lag4, ret1_lag5 |
| **sentiment** | sentiment, sent_mom3, news_spike |

Raw price-level columns (SMA/EMA values) stay in the panel for charts but are never model inputs, because levels are not comparable across tickers; price-to-MA ratios, returns and lagged returns are used instead. Controls against look-ahead: (1) features use only past/current data; (2) the split is chronological by date; (3) sequence windows are built inside each split and each target lies inside its own split, which doubles as an embargo; (4) the scaler is fit on the training split only and the target is scaled separately and inverted for reporting; (5) hyper-parameter search uses an expanding-window walk-forward CV with the scaler refit inside every fold; (6) news after the 16:00 close is mapped to the next session; (7) macro releases respect publication lags; (8) the best model is chosen on validation error, never on the test window.

## 5. Exploratory data analysis

The notebook notebooks/01_eda.ipynb contains every required analysis; each ends in a computed finding that changes a later decision.

- Stationarity: price is non-stationary for 8/10 tickers while returns are stationary for 10/10 (ADF) -> model returns, not price.
- Correlation: mean pairwise return correlation 0.28 (range 0.16 to 0.42) -> diversification is available; use the full shrunk covariance matrix.
- Volatility: correlation of realised volatility with VIX 0.89; lag-1 ACF of squared returns 0.06 -> volatility clusters; keep VIX and rolling-volatility features.
- Distribution: skew 0.02, excess kurtosis 12.4, Jarque-Bera p 0 -> fat tails; report drawdown, empirical and Student-t VaR; Huber loss for deep models.
- Seasonality: month-of-year ANOVA p = 0.302; STL seasonal range 0.112 -> calendar features expected to carry little signal.

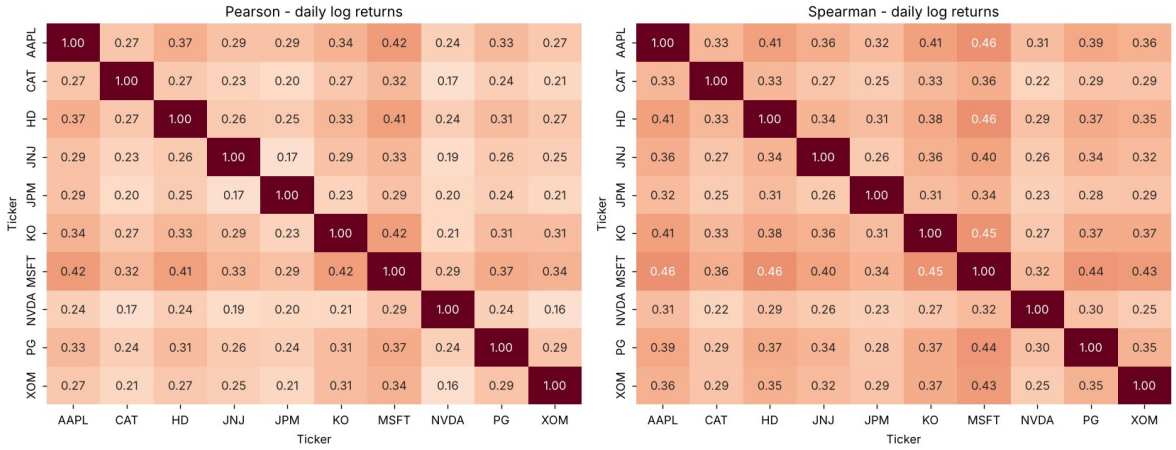



*Figure 1 - Pearson and Spearman return correlations*

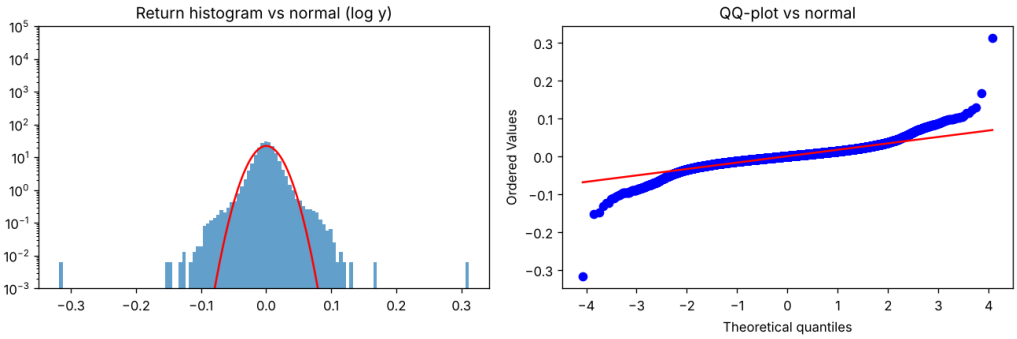



*Figure 2 - Return distribution: fat tails and negative-skew check*

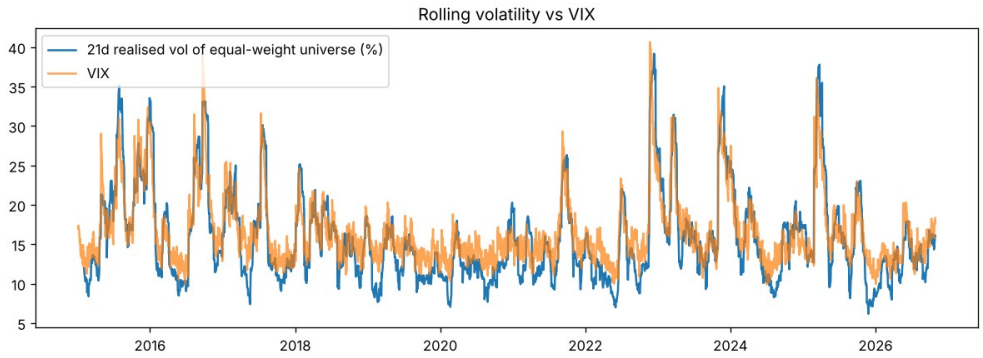



*Figure 3 - Rolling volatility vs VIX*

## 6. Mathematical foundations (from scratch, verified)

src/mathfin.py implements returns, variance, covariance, correlation, Beta, portfolio variance w'Sw, Sharpe, Sortino, maximum drawdown, Calmar, historical and Student-t VaR, a block-bootstrap Sharpe confidence interval, the error metrics, and the manual gradient-descent regression of Listing 7.1. Each is checked against numpy/pandas/scipy/scikit-learn/PyPortfolioOpt.

| Check | Result |
|---|---|
| **simple return** | PASS |
| **log return** | PASS |
| **variance** | PASS |
| **volatility** | PASS |
| **covariance** | PASS |
| **correlation** | PASS |
| **beta** | PASS |
| **cov matrix** | PASS |
| **portfolio vol** | PASS |
| **sharpe** | PASS |
| **max drawdown** | PASS |
| **var hist** | PASS |
| **gd matches ols weights** | PASS |
| **gd matches ols bias** | PASS |
| **cov vs pypfopt sample** | PASS |

Gradient descent converged to w = [1.50802, -2.00596, 0.50409], b = 0.70505; closed-form OLS gives w = [1.50802, -2.00596, 0.50409], b = 0.70505.

## 7. Classical machine-learning models

Linear/Ridge, Random Forest, XGBoost and SVR consume the same engineered feature table and the same chronological samples as the deep models. Each is tuned by grid search over an expanding-window, date-based TimeSeriesSplit (5 folds, embargo = horizon) inside the training split, with the scaler and a separate target scaler inside the pipeline.

| Model_horizon | Selected hyper-parameters | CV MAE (scaled-return units) |
|---|---|---|
| **ridge_h1** | alpha=1000 | 0.01182 |
| **rf_h1** | max_depth=4, min_samples_leaf=50, | 0.01183 |

| | | |
|---|---|---|
| | n_estimators=200 | |
| **xgb_h1** | learning_rate=0.03, max_depth=2, n_estimators=200, subsample=0.8 | 0.01209 |
| **svr_h1** | C=0.1, epsilon=0.1, gamma=scale, kernel=rbf | 0.01182 |
| **ridge_h5** | alpha=1000 | 0.02855 |
| **rf_h5** | max_depth=4, min_samples_leaf=20, n_estimators=200 | 0.02912 |
| **xgb_h5** | learning_rate=0.03, max_depth=2, n_estimators=200, subsample=0.8 | 0.02928 |
| **svr_h5** | C=0.1, epsilon=0.1, gamma=scale, kernel=rbf | 0.02862 |

## 8. Deep-learning models

LSTM, GRU, Bidirectional LSTM and a 2-block Transformer encoder (sinusoidal positional encoding, 4-head attention) are trained on (samples, 60, features) windows with Adam (1e-3), Huber loss, batch 64, shuffle=False, dropout and early stopping (patience 12, best weights restored) on the chronological validation split. All seeds are fixed and TensorFlow op-determinism is enabled.

| Model_horizon | Epochs | Best epoch | Train loss | Val loss | Diagnosis |
|---|---|---|---|---|---|
| **lstm_h1** | 19 | 7 | 0.3479 | 0.3859 | overfitting (validation loss rose while training loss kept falling) |
| **gru_h1** | 18 | 6 | 0.3470 | 0.3879 | overfitting (validation loss rose while training loss kept falling) |
| **bilstm_h1** | 15 | 3 | 0.3475 | 0.3882 | overfitting (validation loss rose while training loss kept falling) |
| **transformer_h1** | 20 | 8 | 0.3508 | 0.3880 | overfitting (validation loss rose while training loss kept falling) |
| **lstm_h5** | 16 | 4 | 0.3911 | 0.4313 | overfitting (validation loss rose while training loss kept falling) |
| **gru_h5** | 16 | 4 | 0.3897 | 0.4311 | overfitting (validation loss rose while training loss kept falling) |
| **bilstm_h5** | 13 | 1 | 0.4055 | 0.4353 | overfitting (validation loss rose while training loss kept falling) |
| **transformer_h5** | 14 | 2 | 0.4065 | 0.4322 | overfitting (validation loss rose while training loss kept falling) |

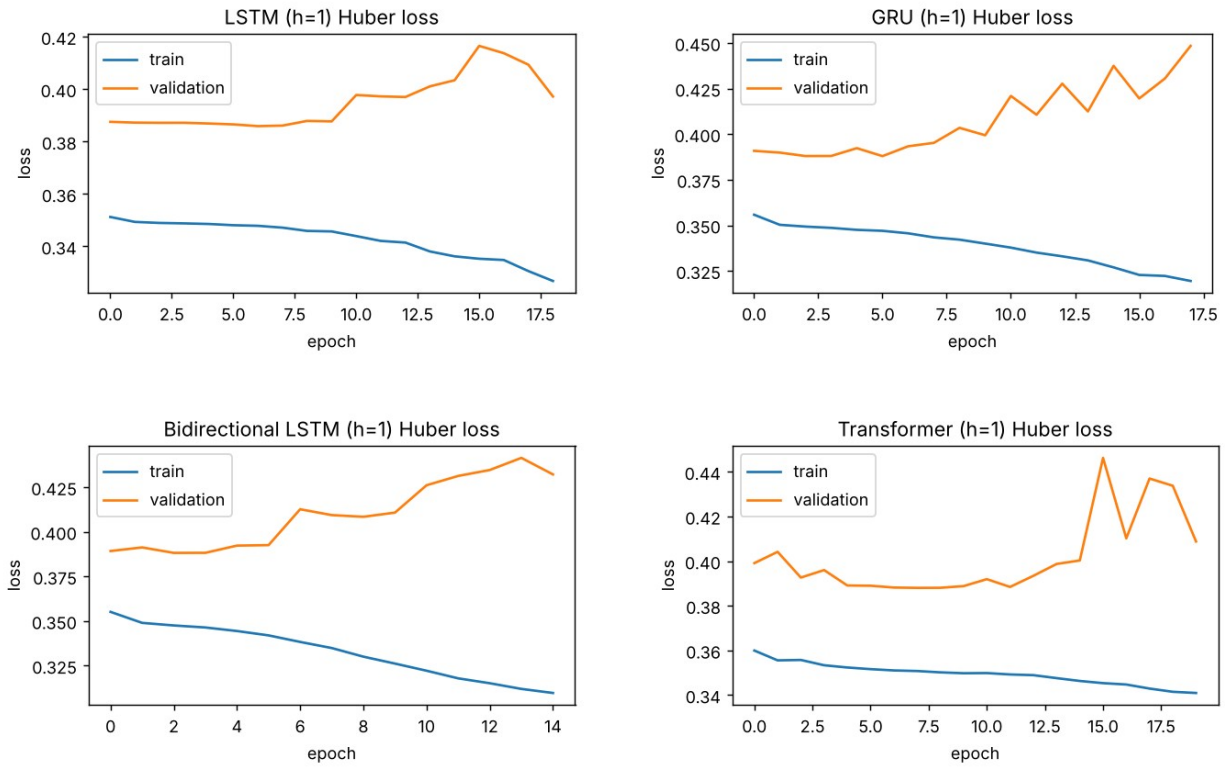



*Figure 4 - Train vs validation loss for each architecture (horizon 1). A widening gap is reported as overfitting, not hidden.*

## 9. Headline result: eight models vs the random walk

Horizon 1 trading day(s) - held-out test window (RMSE/MAE/MAPE/R2 are in price space, averaged over tickers; direction pooled). The random walk predicts today's price; its direction call repeats the trailing move.

| Model | RMSE | MAE | MAPE | R2 | DirAcc | n |
|---|---|---|---|---|---|---|
| **Naive random walk (baseline)** | 6.921 | 4.904 | 1.10 | 0.9642 | 50.6 | 3420 |
| **Linear / Ridge Regression** | 6.908 | 4.892 | 1.11 | 0.9640 | 50.4 | 3420 |
| **Random Forest** | 6.920 | 4.895 | 1.10 | 0.9642 | 51.6 | 3420 |
| **XGBoost** | 6.899 | 4.888 | 1.10 | 0.9642 | 52.2 | 3420 |
| **SVR** | 6.951 | 4.913 | 1.11 | 0.9637 | 51.5 | 3420 |
| **LSTM** | 6.904 | 4.902 | 1.10 | 0.9641 | 51.7 | 3420 |
| **GRU** | 6.923 | 4.910 | 1.10 | 0.9641 | 51.5 | 3420 |
| **Bidirectional LSTM** | 6.929 | 4.916 | 1.10 | 0.9640 | 49.9 | 3420 |
| **Transformer** | 6.917 | 4.907 | 1.10 | 0.9642 | 51.6 | 3420 |

Horizon 5 trading day(s) - held-out test window (RMSE/MAE/MAPE/R2 are in price space, averaged over tickers; direction pooled). The random walk predicts today's price; its direction call repeats the trailing move.

| Model | RMSE | MAE | MAPE | R2 | DirAcc | n |
|---|---|---|---|---|---|---|
| **Naive random walk (baseline)** | 15.187 | 11.428 | 2.59 | 0.8259 | 49.8 | 3380 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Linear / Ridge Regression** | 15.184 | 11.390 | 2.59 | 0.8243 | 52.2 | 3380 |
| **Random Forest** | 15.217 | 11.468 | 2.64 | 0.8180 | 53.1 | 3380 |
| **XGBoost** | 15.159 | 11.410 | 2.62 | 0.8215 | 53.0 | 3380 |
| **SVR** | 15.218 | 11.406 | 2.61 | 0.8224 | 51.8 | 3380 |
| **LSTM** | 15.163 | 11.392 | 2.61 | 0.8234 | 50.0 | 3380 |
| **GRU** | 15.137 | 11.441 | 2.61 | 0.8252 | 52.7 | 3380 |
| **Bidirectional LSTM** | 15.196 | 11.456 | 2.61 | 0.8232 | 49.1 | 3380 |
| **Transformer** | 15.141 | 11.385 | 2.58 | 0.8257 | 53.1 | 3380 |

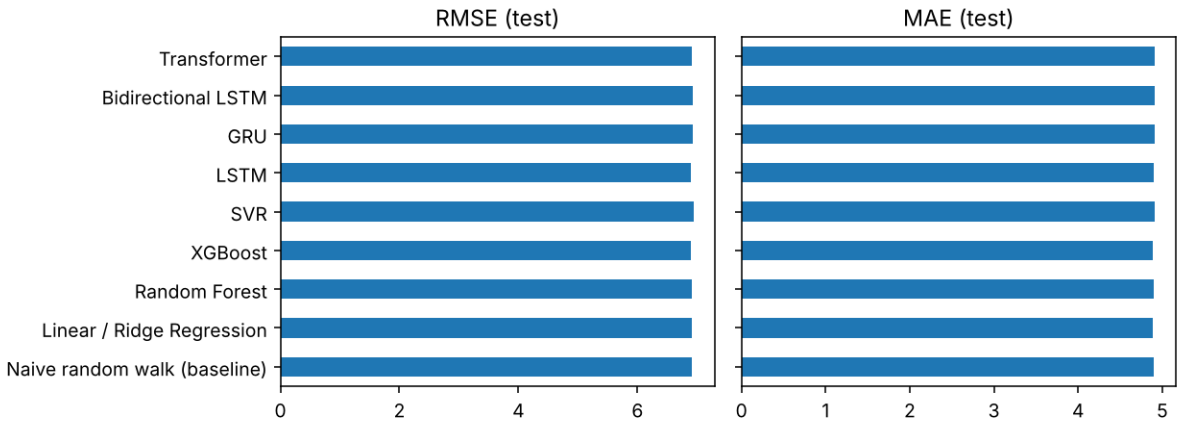



*Figure 5 - RMSE and MAE by model (horizon 1, test)*

Interpretation (h=1). Selected on validation MAE: LSTM. On the test window the lowest MAE is XGBoost (4.888 vs random walk 4.904) and the highest directional accuracy is XGBoost (52.2% vs 50.6%). Best classical model MAE 4.888; best deep model MAE 4.902. The acceptance bar (beat the random walk on MAE and direction) is met, but the MAE margin is only 0.03% - well inside sampling noise, so this should be read as 'on par with the random walk', not as demonstrated skill. R2 in price space is close to 1 for every model including the random walk because price levels are highly autocorrelated; it should not be read as skill. Daily returns are dominated by noise, so on modest daily datasets heavily regularised linear and shallow-tree models usually match or beat deep sequence models (fewer parameters to overfit, and the deep nets' extra capacity has little signal to capture); the loss-curve diagnostics above show where the deep models overfit or fail to learn.

Interpretation (h=5). Selected on validation MAE: GRU. On the test window the lowest MAE is Transformer (11.385 vs random walk 11.428) and the highest directional accuracy is Random Forest (53.1% vs 49.8%). Best classical model MAE 11.390; best deep model MAE 11.385. The acceptance bar (beat the random walk on MAE and direction) is NOT met and is reported as a failure, as the brief requires. R2 in price space is close to 1 for every model including the random walk because price levels are highly autocorrelated; it should not be read as skill. Daily returns are dominated by noise, so on modest daily datasets heavily regularised linear and shallow-tree models usually match or beat deep sequence models (fewer parameters to overfit, and the deep nets' extra capacity has little signal to capture); the loss-curve diagnostics above show where the deep models overfit or fail to learn.

## 10. Sentiment module and its measured effect

Headlines are cleaned (lower-case, strip tickers/URLs, expand abbreviations), scored, mapped to the next session when published after the 16:00 close, and aggregated to a daily mean polarity per asset with a 3-day momentum and a news-volume spike flag. Scorer requested: finbert; scorer used in this run: vader (the FinBERT weights could not be downloaded in the build sandbox, so the VADER lexicon scorer was used; FinBERT is wired in and used automatically when available). 29810 headlines were scored. A market-mood index averages sentiment across the universe (bullish > +0.05, bearish < -0.05).

| LSTM (h=1) | Test MAE | Dir. acc. (%) |
|---|---|---|
| **with sentiment** | 4.9020 | 51.70 |
| **without sentiment** | 4.9074 | 49.97 |
| **difference** | -0.0055 | 1.73 |

Honest conclusion: sentiment improved both MAE and directional accuracy on this dataset (the changes are small relative to run-to-run and sampling noise; one retrained model per arm cannot establish significance). Because this run uses synthetic headlines whose tone merely reacts to same-day returns, no incremental predictive information is expected.

## 11. Portfolio optimisation (Modern Portfolio Theory)

Expected returns: 50% historical mean + 50% model-implied (last 63 validation days), estimated up to the validation end only. This blend is documented and justified as a hedge: historical means are backward-looking and noisy, model-implied returns are forward-looking but shrunk towards zero, and blending limits reliance on either. Covariance: Ledoit-Wolf shrinkage. Constraints: long-only, fully invested, per-asset cap 30%, risk-free rate 4.0%. Monte-Carlo cross-check: the best of 20,000 random portfolios has Sharpe 0.354 vs the optimiser's 0.374 (optimiser on/above the cloud edge).

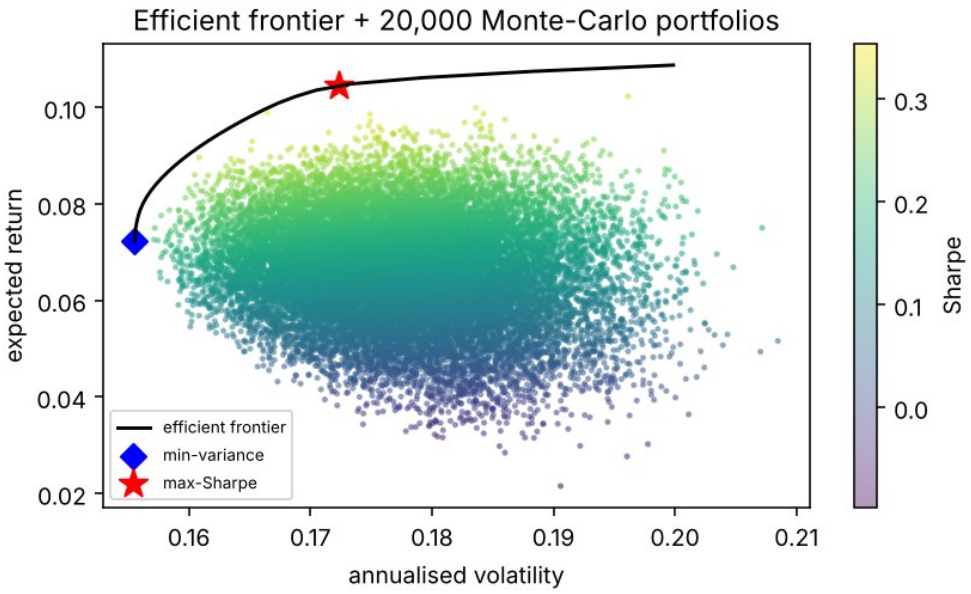



*Figure 6 - Efficient frontier, minimum-variance and max-Sharpe portfolios, Monte-Carlo cloud*

Max-Sharpe weights: HD 30.0%, NVDA 25.8%, JPM 20.6%, CAT 15.3%, MSFT 8.3%.

| OOS portfolio | Ann. ret | Ann. vol | Sharpe | Sortino | Beta | Max DD | Calmar | VaR95 hist | VaR95 t |
|---|---|---|---|---|---|---|---|---|---|
| **Optimised** | 0.158 | 0.159 | 0.750 | 1.107 | 0.936 | -0.101 | 1.559 | -0.015 | -0.016 |
| **Equal-weight** | 0.139 | 0.147 | 0.686 | 1.013 | 1.063 | -0.090 | 1.537 | -0.014 | -0.014 |
| **Benchmark** | 0.126 | 0.123 | 0.701 | 1.038 | 1.000 | -0.082 | 1.529 | -0.012 | -0.012 |

Backtest (weights frozen at the end of validation, daily rebalanced to constant weights on the untouched test window): optimised Sharpe 0.75 [-0.56, 1.98] vs equal-weight 0.69 [-0.53, 1.94] and benchmark 0.70 (block-bootstrap 95% CIs). The acceptance bar (optimised Sharpe > equal-weight) is met, but the overlapping confidence intervals show the difference is not statistically established.

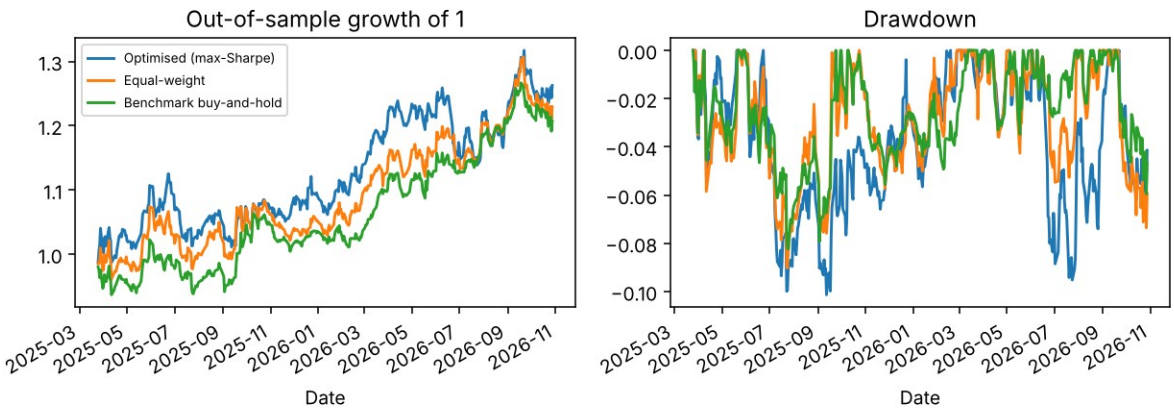



*Figure 7 - Out-of-sample growth and drawdown*

## 12. Recommendation engine

Composite = 1.0 x forecast + 0.5 x sentiment - 0.5 x risk, each sub-signal z-scored across the universe per date; the risk signal averages z-scores of 21-day volatility, 252-day Beta and drawdown from the 252-day high. BUY if composite >= 0.5, SELL if <= -0.5, otherwise HOLD (a no-trade band). Rebalancing flags any asset whose target weight differs from the current weight by more than 5% (current holdings assumed equal-weight). Every recommendation is shown with its three sub-scores in the dashboard.

| Horizon | Model | BUY calls | BUY hit-rate | SELL calls | SELL hit-rate | Buy-and-hold up-rate |
|---|---|---|---|---|---|---|
| **1** | LSTM | 1052 | 50.7% | 1163 | 46.5% | 51.3% |
| **5** | GRU | 1101 | 52.2% | 1123 | 45.6% | 53.1% |

Hit-rate = share of calls followed by a forward return of the called sign on the test window; the buy-and-hold up-rate is the unconditional share of positive forward returns.

## 13. Dashboard and reproducibility

dashboard/app.py is a single Streamlit application with the seven required panels (Overview, Price & prediction, Model comparison, Portfolio analytics, Risk dashboard, Sentiment, Recommendations), Plotly charts with hover detail, controls for ticker, horizon and risk-free rate, cached loading of pre-computed outputs (it never retrains) and a persistent educational-use disclaimer. Reproducibility: config.yaml holds the universe, dates, split ratios, sequence length and seed; requirements.txt is pinned; `make data` rebuilds the dataset, `make train` retrains and re-evaluates everything, `make report` regenerates the notebook and this report.

## 14. Limitations and honest failure analysis

- Synthetic data in this build: all reported figures illustrate the pipeline, not real-market skill. A single hold-out test window (about 1.7 years) is one market regime; results can differ materially in other regimes.
- Forecasting: next-day returns are close to unpredictable; the horizon-1 acceptance bar is met, and horizon 5 is not met. Differences between models are small relative to sampling noise; no formal significance test (e.g. Diebold-Mariano) is run.
- Price-space R2 is inflated by price autocorrelation and is not evidence of skill.
- The model-implied expected return is a noisy, shrunk estimate; mean-variance weights are sensitive to mu, which is why the cap, Ledoit-Wolf shrinkage and a historical blend are used.
- The backtest ignores transaction costs, slippage and taxes, and assumes daily rebalancing to constant weights.
- Sentiment: free news APIs provide roughly one year of history; the FinBERT scorer needs a model download; VADER is a general-purpose lexicon and mislabels some finance text.
- Survivorship bias: the universe is today's large-cap leaders; this flatters historical returns on real data.
- Dividend adjustments are taken from the vendor's adjusted close and cannot be independently verified.

## Appendix A - Deliverables checklist (PRD 19)

- Git-ready repository with the layered structure of Listing 15.1 and pinned requirements - done
- Deterministic ingestion + cleaning scripts and a data-quality report (reports/results/data_quality_report.md) - done
- Feature engineering and sequence-generation modules with no look-ahead - done (src/features.py)
- EDA notebook with captioned, decision-linked findings - done (notebooks/01_eda.ipynb)
- From-scratch financial-maths module verified against libraries - done (src/mathfin.py)
- Four tuned ML models and four DL models with the populated eight-model table - done (Section 9)
- Sentiment pipeline and the with/without-sentiment result - done (Section 10)
- Efficient-frontier optimisation, max-Sharpe weights and baseline-relative backtest - done (Section 11)
- Transparent recommendation and rebalancing engine - done (src/recommend.py)
- Streamlit dashboard with all seven panels - done (dashboard/app.py)
- Written report and README with one-command rebuild instructions - done

*Educational research and decision-support tool only. Outputs are not financial advice.*